

PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

Team 34: The Hacktivist

Silver Finn

1. Project Overview

- **Project Name:** Silver Finn - Intelligent Workshop Management Platform.
 - **Problem Statement:** Auto workshops lack a cohesive, intelligent interface for mechanics to log inspections, access technical knowledge, and predict maintenance needs based on historical data.
 - **Target Domain:** Automotive Service Industry and Workshop Management.
 - **Proposed Solution Summary:** A mechanic-facing iPad web application that integrates a cinematic 3D car viewer (Asphalt 8/9 style) with AI-powered diagnostics and proactive maintenance alerts.
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2. Background & Business Objective

- **Background:** Traditional workshop management is often manual or lacks visual depth. Silver Finn leverages existing 3D models (e.g., Mercedes C63 AMG) to create an interactive "digital twin" for vehicle inspections.
 - **Importance:** By transforming the mechanic's role from manual logger to a "curator" assisted by AI, the platform accelerates service times and increases diagnostic accuracy.
 - **Strategic Fit:** Designed for hackathon execution (UMHack) with a focus on high-fidelity visual engagement and AI-native features like RAG (Retrieval-Augmented Generation).
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3. Product Purpose

3.1. Main Goal

To provide an intelligent, 3D-interactive partner for mechanics and service advisors to manage vehicle health and customer history.

3.2. Intended Users

- **Mechanics:** Primary users for checklists and 3D interaction.
 - **Service Advisors:** Manage customer profiles and service sessions.
 - **Admins:** Manage workshop settings and multi-franchise data.
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4. System Functionalities

4.1. Key Functionalities

- **3D Guided Checklist:** An interactive flow where 3D hotspots pulse to guide mechanics through service zones (Engine, Brakes, Tyres, etc.).
- **RAG Chat Assistant:** A context-aware AI assistant that answers technical questions (e.g., torque specs) using a knowledge base of OEM manuals.
- **Predictive Maintenance:** A statistical engine that analyzes ~750–1000 records to predict part failures based on mileage and model history.
- **Service History & Timeline:** Comprehensive logs of past sessions, including replaced parts, costs, and mechanic notes.

4.2. AI Model & Tech Stack

Layer	Technology
LLM / RAG	Gemini
Vector DB	ChromaDB (self-hosted in-process)
Embeddings	Gemini Embedding API
Frontend	React 19 + Vite 8 + Three.js + R3F
Backend	Node.js + Express with Prisma ORM
Database	PostgreSQL on Neon (free tier)

5. User Stories & Use Cases

- **Mechanic:** "As a mechanic, I want to be guided through the vehicle zones via 3D hotspots so that I don't miss any critical inspection points".
 - **Service Advisor:** "As a service advisor, I want to see a probability-based alert for brake pads so that I can proactively suggest replacements to the customer".
 - **Prompt-to-Knowledge Flow:** Mechanic types a query then Backend embeds query via Gemini then ChromaDB retrieves top 5 manual chunks then Gemini generates response.
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6. Features Included (Scope)

- **Guided Hotspot Mode:** 8 defined zones (Engine, Front-Left, Undercarriage, etc.) with PASS/FAIL toggles.
 - **Photo Documentation:** Upload photos of replaced parts directly to Cloudinary.
 - **PDF Export:** Generate printable client-side service reports using jsPDF.
 - **Multi-tenant Support:** Database schema designed for multi-workshop/franchise readiness.
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7. Features Not Included

- **Customer-facing App:** The current scope is strictly internal/mechanic-facing.
 - **Live Database AI Access:** RAG is limited to the knowledge base, not live transactional DB data.
 - **Automatic VIN/Plate Recognition:** Car registration is currently manual.
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8. Implementation Plan

- **Phase 1:** Backend foundation, PostgreSQL/Prisma setup, and Prediction API.
 - **Phase 2:** Dark metallic theme transformation and 5-tab core layout.
 - **Phase 3:** 3D Guided Checklist with pulsing hotspots and item cards.
 - **Phase 4:** Service history timeline and PDF export functionality.
 - **Phase 5:** RAG Chat implementation with Gemini and ChromaDB.
 - **Phase 6:** Analytics dashboard and proactive alert generation.
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9. Risks & Open Questions

- **Data Accuracy:** The prediction engine relies on ~750 fake records; real-world accuracy may vary.
- **Zone Completion:** Whether mechanics can skip zones (N/A) or must follow a strict order is TBD.
- **Hardware Assets:** Usage of existing C63 AMG model as a demo vs. adding specific Malaysian models (Proton/Perodua) later.